

CipherSight: AI-Powered Phishing Detection

Hello team! Today I'm excited to present **CipherSight**, our robust, end-to-end phishing detection platform. Built with security and scalability at its core, CipherSight separates highly sophisticated spoofed domains from legitimate enterprise properties.

1. The Backend Engine

CipherSight's backend is a high-performance **FastAPI** application, completely secured using JWT tokens and strict Role-Based Access Control (RBAC). Data persistence is handled reliably by **PostgreSQL** with Alembic keeping our schemas perfectly migrated.

To ensure safety and reliability, we've implemented strict SSRF (Server-Side Request Forgery) isolation and sandboxed crawlers. We never ping live phishing sites during testing. Our comprehensive, 100% passing pytest suite (19/19) uses offline fixtures.

Pipeline Stages

Our pipeline analyzes domains through sequential layers of defense: **DOMAIN ? WHOIS/RDAP ? URL Analysis ? HTML/Content ? Screenshot ? Visual Similarity ? AI Risk Engine ? Final Verdict**

Once analyzed, reports can be seamlessly exported to JSON, CSV, and PDF!

2. The Frontend Experience

The user interface is a clean, responsive Dashboard separated into two main views:

- **Home Page:** A quick-scan view where users submit domains and see top-level risk metrics.

- **Dashboard:** A detailed SOC intelligence view displaying the complete pipeline, historical data, and deep visual similarity analytics.

Verdicts are clearly classified as **SAFE**, **HIGH RISK**, or **NEEDS REVIEW**. *(Note: We recently resolved a minor data-mapping issue on the Home page, so all JSON fields like domain_age and visual_match now sync perfectly with the backend!)*

3. Integration & Data Flow

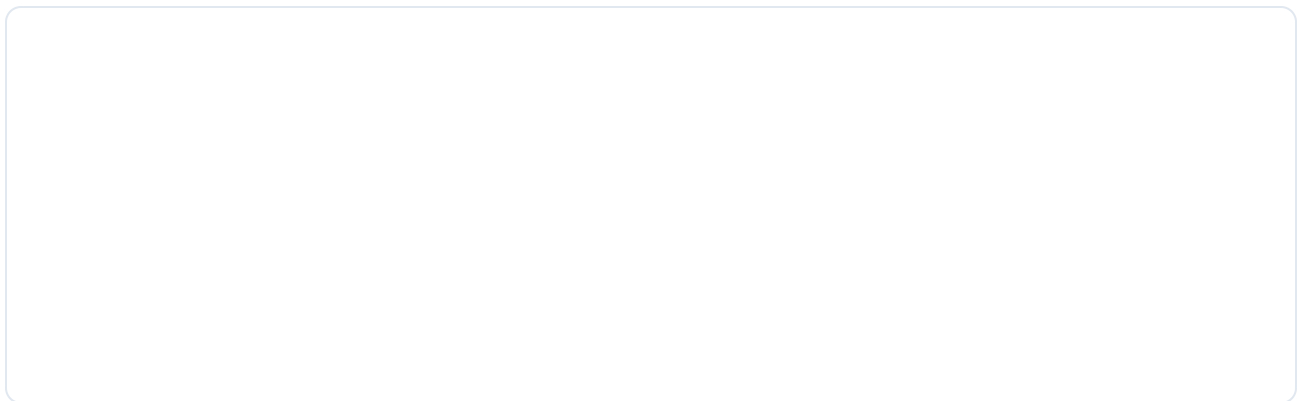
The frontend communicates seamlessly with our FastAPI backend via RESTful endpoints (/scans, /verdict, /brands).

Impact

When an analyst submits a domain, they are instantly presented with an Explainable AI risk score. This transparent pipeline allows us to easily differentiate a real domain (like sbi.bank.in) from a dangerous lookalike by fusing signals across domain age, entropy, endpoint risk, and visual clones.

Summary: The system is feature-complete, secure, and ready for deployment. The frontend and backend contracts are perfectly synced!

System Architecture & Integration Flow



The Multi-Layered Defense Engine

flowchart LR
S((Scan Target)) --> K{Kill Switch}
K -- "< 30 days old + Brand Match" --> CRIT[CRITICAL]
K -- "Passes" --> H{Hard Signals}
H -- "SSL / Privacy / Entropy" --> HIGH[HIGH RISK]
H -- "Passes" --> SF[Soft Signals]
SF -- "Visual and Text Similarity" --> SAFE[SAFE / NEEDS REVIEW]